

TOMTOM

Tactical Overwatch Module for Tracking, Observation and Mapping

Concept of Operations (CONOPS)

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1. Introduction

1.1 Purpose

The purpose of this Concept of Operations (CONOPS) is to describe the intended operational use, mission objectives, stakeholders, and high-level capabilities of the Tactical Overwatch Module for Tracking, Observation and Mapping (TOMTOM). This document establishes the operational context for the system and provides a foundation for requirements development, system architecture, verification planning, and flight-test activities.

1.2 Scope

TOMTOM is a mission systems demonstrator designed to provide intelligence, surveillance, reconnaissance, tracking, observation, and mapping capabilities through the integration of airborne, ground-based, and operator-facing system components. The system is intended to support situational awareness, information collection, target observation, and mission planning by gathering, processing, and disseminating operationally relevant information. This document describes the operational concept of TOMTOM and establishes the framework for requirements development, system architecture, verification planning, and system testing.

1.3 References

Reference ID/ Reference

[R1] International Council on Systems Engineering (INCOSE). INCOSE Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities (5th ed.). Wiley, 2023.

[R2] Federal Aviation Administration. 14 CFR Part 107 – Small Unmanned Aircraft Systems. U.S. Department of Transportation.

[R3] Object Management Group. OMG Systems Modeling Language (SysML®) Version 1.7. Formal Specification, 2024.

[R4] Dassault Systèmes. Cameo Systems Modeler User Documentation.

[R5] QGroundControl Development Team. QGroundControl User Guide and Documentation.

[R6] ArduPilot Development Team. ArduPilot Documentation.

[R7] TAK Product Center. Android Team Awareness Kit (ATAK) Documentation.

[R8] OpenTAKServer Development Team. OpenTAKServer Documentation.

1.4 Acronyms, Abbreviations, and Definitions

1.4.1 Acronyms

CONOPS – Concept of Operations

ISR – Intelligence, Surveillance, and Reconnaissance

MBSE – Model-Based Systems Engineering

MOE – Measure of Effectiveness

MOP – Measure of Performance

SysML – Systems Modeling Language

TOMTOM – Tactical Overwatch Module for Tracking, Observation and Mapping

UAS – Unmanned Aircraft System

1.4.2 Abbreviations

1.4.3 Definitions

Mission Thread – A sequence of operational activities that accomplishes a mission objective.

Stakeholder – An individual or organization that affects or is affected by the system.

Operational Scenario – A representative sequence of events describing system use.

2. Mission Need

2.1 Problem Statement

Small teams and operators often have limited ability to rapidly collect, organize, and distribute information about areas of interest beyond their immediate line of sight. Existing methods may require significant manpower, expose personnel to unnecessary risk, or delay decision-making. As modern operations increasingly depend upon timely, accurate, and accessible information, the inability to rapidly gather and integrate data from multiple sources can reduce situational awareness and operational effectiveness. A need exists for a flexible mission system capable of extending situational awareness through the collection, processing, integration, and dissemination of operationally relevant information from airborne, ground-based, and operator-facing system components.

2.2 Mission Context

Modern defense, emergency response, and security operations frequently require personnel to obtain information from environments that are difficult, hazardous, or

impractical to access directly. These environments may include urban structures, confined indoor spaces, damaged infrastructure, remote outdoor locations, and areas with limited visibility or restricted access.

TOMTOM is intended to provide a lightweight, portable, and rapidly deployable mission system capable of extending situational awareness beyond the operator's immediate line of sight. The system is designed to support information collection, observation, tracking, mapping, and area characterization activities through the integration of airborne, ground-based, and operator-facing system components.

The system is envisioned for employment in support of defense-oriented missions, including reconnaissance, route assessment, force protection, and operational planning. The same capabilities may also support non-defense applications such as search and rescue, disaster response, emergency management, infrastructure inspection, and environmental assessment.

TOMTOM is intended to operate across a range of environments, including both indoor and outdoor settings. The system emphasizes portability, rapid deployment, modularity, and ease of operation to enable effective use in dynamic operational conditions where timely information is critical to decision-making.

2.3 Desired Operational Outcomes

The desired outcome of TOMTOM is to provide operators with timely, relevant, and actionable information that improves situational awareness and supports informed decision-making. Successful employment of the system should enable personnel to better understand, monitor, and characterize areas of interest while reducing the need for direct exposure to potentially hazardous or inaccessible environments.

The system should support the rapid collection, processing, and dissemination of information from both indoor and outdoor operating environments. Information provided by TOMTOM should assist operators in identifying points of interest, tracking observed activity, assessing terrain and structures, and developing an improved understanding of the operational environment.

Through its lightweight, portable, and rapidly deployable design, TOMTOM should enable users to establish an organic information collection capability with minimal setup time and logistical burden. The system should support a variety of mission types, including defense, security, emergency response, search and rescue, and infrastructure assessment activities.

3. Stakeholders

Stakeholder	Category	Role
Serena Green	Project	Student Engineer
Jesse Dimas	Project	Technical Mentor
System Operator	Operational	End User
Mission Lead	Operational	Decision Maker
System Maintainer	Operational Support	Support Personnel
Department of Defense Organizations	Potential End User	Defense Stakeholder
Emergency Response Organizations	Potential End User	Emergency Services Stakeholder
Public Safety Organizations	Potential End User	Security Stakeholder
Infrastructure Assessment Organizations	Potential End User	Civil Stakeholder
Federal Aviation Administration (FAA)	Regulatory	Aviation Authority
Local Airspace Authorities	Regulatory	Airspace Authority

4. Operational Environment

4.1 Assumptions

The following assumptions are made for the development and operation of TOMTOM:

1. Operators will receive basic training on system deployment, operation, recovery, and emergency procedures.
2. Mission objectives and operating areas will be defined before system deployment.
3. Required system components will be available, charged, inspected, and functional at the start of the mission.
4. Outdoor operations may use GPS when available; however, TOMTOM is intended to support degraded-GPS or GPS-denied operation through onboard

- sensing, local mapping, visual reference, inertial sensing, or operator-assisted navigation.
5. Indoor or confined space operations may require operator-assisted navigation, visual positioning, onboard sensing, or local reference points due to limited GPS availability.
 6. Communications links are expected to remain available within the intended operating area, though signal quality may vary based on terrain, structures, and distance.
 7. Environmental conditions are expected to remain within the safe operating limits of system components.
 8. TOMTOM is intended to support low-battery response behavior, including return-to-operator, return-to-charge, docking, landing, or safe shutdown depending on the mission phase and operating environment.
 9. Regulatory approval or operational authorization for flight activities will be obtained when required.
 10. Commercially available components will be used whenever practical to support project feasibility, affordability, and schedule constraints.

4.2 Constraints

The design and operation of TOMTOM are subject to the following constraints:

1. TOMTOM must comply with applicable FAA regulations and local operating restrictions governing unmanned aircraft operations.
2. System portability is constrained by the need for single-operator transport, setup, deployment, and recovery.
3. System endurance is constrained by available battery capacity, payload weight, operating mode, and environmental conditions.
4. Autonomous recharge capability is constrained by docking accuracy, charging hardware, landing environment, battery safety, system weight, and project schedule.
5. GPS-denied operation may reduce navigation accuracy, mapping precision, tracking performance, and autonomy capability.
6. Indoor and confined space operations may be constrained by obstacle density, lighting conditions, communications interference, limited maneuvering space, and reduced GPS availability.
7. Communications performance may be constrained by terrain, structures, distance, interference, and antenna placement.
8. Payload capacity is constrained by the size, lift capability, and weight limits of the airborne platform.

9. Weather conditions, including wind, precipitation, temperature, and visibility, may limit outdoor deployment and mission effectiveness.
10. Available project budget, development time, technical resources, and student project scope constrain system complexity and capability growth.
11. Safety considerations constrain system operation around people, property, aircraft, vehicles, and sensitive areas.
12. Initial TOMTOM development will prioritize demonstrable mission capability, requirements traceability, MBSE artifacts, and test documentation over full operational maturity.

5. Operational Concept

5.1: Mission Objectives

1. Extend situational awareness beyond operator line of sight.
2. Collect and disseminate operationally relevant information.
3. Support observation, tracking, and mapping activities.
4. Operate in both indoor and outdoor environments.
5. Support operation in GPS-degraded or GPS-denied conditions.
6. Minimize operator exposure to hazardous environments.
7. Enable rapid deployment and recovery.
8. Support modular payload integration.
9. Provide persistent information collection through battery management and autonomous recharge capabilities.
10. Maintain portability suitable for single operator use.

5.2 Concept of Employment

TOMTOM is intended to be employed as a mission agnostic, portable mission systems capability supporting reconnaissance, observation, tracking, and mapping activities in environments where direct observation is impractical, hazardous, or impossible. The system is designed for operation by a small team or single operator and emphasizes rapid deployment, ease of use, and portability.

Prior to mission execution, the operator defines the mission objective, area of interest, operating constraints, and desired information products. The operator deploys the airborne platform, ground control components, and mission management interface from a portable transport configuration.

Following system initialization and mission checks, the airborne platform is launched to collect information within the designated operating area. Sensor data is gathered through onboard payloads and transmitted to the operator. Information may include video, imagery, location data, maps, observations, tracked objects, and other mission-relevant information.

During mission execution, the system supports observation, mapping, target tracking, route assessment, and area characterization activities. Information collected by the system is processed and presented to the operator to support situational awareness and decision-making.

If battery levels become low, communication links are degraded, or mission objectives are completed, the system executes predefined recovery procedures. Recovery actions may include return-to-operator, return-to-charge, autonomous docking, controlled landing, or mission termination depending on the operational context.

Upon mission completion, collected video, imagery, telemetry, Cursor-on-Target events, and mission metadata are stored in a cloud-based mission data repository for review, analysis, and traceability. When connected to a TAK environment, TOMTOM may publish position, observation, and tracking data to a TAK Server for display in ATAK-compatible clients.

5.3 Mission Threads

5.3.1 MT-001 Nominal ISR Mission

Step	Actor	Activity	Output
1	Mission assigned to Mission Lead	Defines area of interest and information requirements	Mission assigned
2	Operator	Deploys TOMTOM system	System operational
3	Operator	Performs pre-mission checks	System verified
4	Operator	Launches airborne platform	Vehicle airborne
5	TOMTOM	Navigates toward area of interest	Area reached
6	TOMTOM Sensors	Collect video, imagery, telemetry, and environmental data	Raw mission data
7	Mission Processor	Processes and organizes collected information	Situational picture

8	TAK Integration	Publishes observations and position updates to ATAK	Shared operational view
9	Operator	Reviews information and identifies points of interest	Assessment generated
10	Mission Lead	Makes operational decision based on collected information	Decision support provided
11	TOMTOM	Returns to operator or docking station	Recovery initiated
12	Mission Archive	Stores video, imagery, telemetry, CoT events, and metadata	Mission archived
13	Operator	Reviews mission products	Mission complete

5.3.2 MT-002 Lost Link Recovery

Step	Actor	Activity	Output
1	Operator	Conducts ISR mission	Mission In progress
2	Communications Subsystem	Detects loss of command and control link	Link loss detected
3	TOMTOM Mission Management	Verifies loss exceeds predefined timeout threshold	Lost link condition confirmed
4	TOMTOM Mission Management	Activates lost link recovery mode	Recovery mode initiated
5	Air Vehicle	Executes predefined recovery procedure	Recovery path established
6	Navigation Subsystem	Navigates toward recovery location or operator position	Recovery transit in progress
7	Operator	Attempts to reestablish communications	Link recovery attempted
8	Communications Subsystem	Reestablishes link or remains disconnected	Link status updated
9	Air Vehicle	Continues recovery or resumes mission based on system logic	Mission disposition determined

10	Air Vehicle	Lands at designated recovery location or returns to operator	Vehicle recovered
11	Mission Data Repository	Archives telemetry, event logs, and mission data	Lost link event recorded
12	Operator	Reviews recovery event and mission data	Mission complete

5.3.3 MT-003 Low Battery Autonomous Recovery

Step	Actor	Activity	Output
1	Operator	Conducts ISR mission	Mission in progress
2	Power Management Subsystem	Monitors battery state of charge	Battery status available
3	Power Management Subsystem	Detects battery level below recovery threshold	Low battery condition detected
4	TOMTOM Mission Management	Evaluates mission status, distance from recovery location, and remaining energy reserves	Recovery decision generated
5	TOMTOM Mission Management	Initiates low battery recovery procedure	Recovery mode activated
6	Air Vehicle	Terminates or suspends active mission tasks	Mission Suspended
7	Navigation Subsystem	Navigates toward operator, charging station, docking station, or designated recovery area	Recovery transit initiated
8	Air Vehicle	Lands or docks at recovery location	Vehicle recovered
9	Charging Subsystem	Begins autonomous battery charging sequence	Charging initiated
10	Mission Management Subsystem	Monitors charging progress and charging sequence	Readiness status available

11	Operator	Reviews mission status and determines whether to resume operations	Mission disposition determined
12	Mission Data Repository	Archives telemetry, battery events, and mission data	Recovery event recorded

5.3.4 MT-004 Target Tracking

Step	Actor	Activity	Output
1	Mission Lead	Defines tracking objective and area of interest	Tracking mission assigned
2	Operator	Deploys and launches TOMTOM	System operational
3	Air Vehicle	Navigates to designated operating area	Area reached
4	Sensor Payload	Detects potential target of interest	Candidate target identified
5	Operator	Confirms target selection	Target established
6	Mission Management Subsystem	Initiates tracking mode	Track established
7	Sensor Payload	Continuously observes target position and movement	Tracking data collected
8	Mission Processing Subsystem	Correlates observations and updates target track	Track maintained
9	TAK Integration Subsystem	Publishes target position and track updates to ATAK	Shared operational picture
10	Operator	Monitors target activity and reviews updates	Situational awareness maintained
11	Mission Lead	Makes operational decisions based on tracking information	Decision support provided
12	Mission Management Subsystem	Detects mission completion, target loss, or operator termination command	Tracking terminated

13	Air Vehicle	Returns to operator or recovery location	Recovery initiated
14	Mission Data Repository	Archives tracking data, observations, telemetry, and mission metadata	Mission archived

6. Operational Scenarios

6.1 MT-001 Nominal ISR Mission

A Mission Lead identifies an area of interest requiring additional observation and situational awareness. Information requirements are established, including desired observations, points of interest, and mission constraints. The Operator deploys TOMTOM from its transport configuration and performs pre-mission checks to verify system readiness, communications connectivity, sensor functionality, and mission configuration.

Following successful verification, the airborne platform is launched and navigates toward the designated operating area. During transit and mission execution, onboard sensors collect video, imagery, telemetry, and environmental information relevant to the mission objective. Collected data is processed by TOMTOM mission systems and organized into a usable operational picture.

Mission information is transmitted to operator-facing interfaces and shared through TAK-compatible systems to support situational awareness and collaborative decision-making. The Operator reviews incoming information, identifies points of interest, and monitors mission progress. Observations and mission products are made available to the Mission Lead to support assessment of the operational environment and inform subsequent decisions.

Upon completion of information collection activities, the airborne platform returns to the Operator, designated recovery location, or charging station. Mission data, including video, imagery, telemetry, observations, Cursor-on-Target events, and associated metadata, is archived within the mission data repository for future review and analysis. The mission is considered successful when TOMTOM provides timely, relevant, and actionable information that improves situational awareness while minimizing the need for personnel to directly enter or observe the area of interest.

6.2 MT-002 Lost Link Recovery

During execution of an ISR mission, TOMTOM is operating within a designated area of interest while collecting and transmitting mission data to the Operator. While the mission is in progress, the communications subsystem experiences a loss of command-and-control connectivity due to distance, terrain masking, interference, environmental conditions, or other factors affecting communications performance.

The communications subsystem detects the interruption and notifies the mission management subsystem of the lost-link condition. TOMTOM verifies that the communication outage exceeds a predefined timeout threshold to distinguish a true lost-link event from temporary signal degradation. Once the condition is confirmed, the system initiates a predefined lost-link recovery procedure.

Depending on mission configuration and operational requirements, recovery actions may include returning to the operator, navigating to a designated recovery location, proceeding to a charging station, executing a controlled landing, or performing another approved recovery behavior. During recovery, onboard navigation and mission management functions continue to monitor vehicle status and support safe operation of the airborne platform.

If communications are successfully re-established during recovery, the system may provide updated status information to the Operator and resume normal operations if authorized. If communications are not restored, TOMTOM continues executing the recovery procedure until the vehicle reaches the designated recovery location and safely terminates flight operations.

Following recovery, mission telemetry, system status information, lost-link events, and mission data are archived within the mission data repository. The Operator reviews recovery information and mission products to assess system performance and determine readiness for future operations.

The scenario is considered successful when TOMTOM safely responds to the loss of communications, preserves system integrity, and recovers the airborne platform without requiring direct operator intervention.

6.3 MT- 003 Low Battery Autonomous Recovery

During execution of an ISR mission, TOMTOM is operating within a designated area of interest while collecting, processing, and disseminating mission information. Throughout mission execution, the power management subsystem continuously monitors battery state of charge and evaluates available energy reserves against mission requirements and recovery needs.

As battery capacity decreases, the power management subsystem determines that the remaining energy reserves have fallen below a predefined recovery threshold. The mission management subsystem evaluates current mission status, vehicle position, distance to available recovery locations, and available battery reserves before initiating a low-battery recovery procedure.

Once the recovery condition is confirmed, TOMTOM notifies the Operator and transitions from mission execution to recovery operations. Active mission tasks are terminated or suspended, and the airborne platform navigates toward a designated recovery location. Depending on system configuration and mission requirements, recovery locations may include the Operator's position, a designated landing area, or an autonomous charging station.

Upon arrival at the recovery location, the airborne platform performs a controlled landing or docking procedure. If autonomous charging capability is available, the charging subsystem initiates battery replenishment and continuously monitors charging status, battery health, and system readiness. Mission management functions maintain awareness of vehicle status and provide updates to the Operator throughout the recovery process.

Following recovery, telemetry, battery status information, charging events, and mission data are archived within the mission data repository for review and analysis. The Operator evaluates mission status and determines whether operations should be terminated, resumed, or re-tasked based on mission objectives and system readiness.

The scenario is considered successful when TOMTOM detects low battery conditions with sufficient margin to safely recover the airborne platform, preserve mission data, and prepare the system for future operations without loss of the vehicle or mission-critical information.

6.4 MT- 004 Target Tracking Scenario

During mission planning, the Mission Lead identifies a target, object, vehicle, person, or point of interest requiring continued observation. The Operator receives tracking objectives, operating constraints, and desired information requirements before deploying TOMTOM and conducting pre-mission system checks.

Following successful system verification, the airborne platform is launched and navigates toward the designated area of interest. Upon arrival, onboard sensors begin collecting video, imagery, telemetry, and observational data. The Operator or mission management functions identify and designate a target for tracking based on mission objectives.

Once a target has been identified, TOMTOM transitions into tracking operations. The sensor payload continuously observes the target while mission processing functions correlate observations and maintain an updated track. Target position, movement,

observations, and associated metadata are organized into a continuously updated operational picture and distributed to operator-facing interfaces.

Tracking information is shared through TAK-compatible systems to support situational awareness and collaborative decision-making. The Operator monitors target activity, reviews tracking information, and identifies significant changes in movement, behavior, or location. The Mission Lead uses the information provided by TOMTOM to support operational planning, decision-making, and mission execution.

Tracking operations continue until mission objectives are achieved, the target is lost, system limitations require mission termination, or the Operator commands tracking to end. Upon completion of the mission, the airborne platform returns to the Operator, designated recovery location, or charging station. Tracking data, observations, telemetry, Cursor-on-Target events, and mission metadata are archived within the mission data repository for future review and analysis.

The scenario is considered successful when TOMTOM maintains awareness of the designated target, provides timely and accurate tracking information to mission personnel, and supports informed decision-making throughout the duration of the mission.

7. System Capabilities

7.1 Mission Management Capabilities

Capability ID	Capability	Description
CAP-100	Mission Management	Coordinate mission planning, execution, monitoring, and recovery activities
CAP-110	Mission Planning	Support mission configuration, objectives, and operational constraints
CAP-120	Mission Monitoring	Monitor mission status, system health, and mission progress
CAP-130	Mission Recovery	Manage recovery actions during mission completion or contingency events
CAP-140	Mission Configuration	Configure mission parameters, payload settings, and operational constraints prior to mission execution.

7.2 ISR Operations Capabilities

Capability ID	Capability	Description
CAP-200	Information Collection	Collect operationally relevant information from the environment
CAP-210	Video Collection	Capture live video from airborne payloads
CAP-220	Imagery Collection	Capture still imagery and visual observations
CAP-230	Telemetry Collection	Collect vehicle and mission telemetry
CAP-240	Environmental Observation	Collect environmental and situational data
CAP-250	Sensor Payload Management	Control and manage onboard sensor payloads during mission operations.

7.3 Situational Awareness Capabilities

Capability ID	Capability	Description
CAP-300	Situational Awareness Generation	Organize collected information into usable mission products
CAP-310	Operator Information Display	Present mission information to operators
CAP-320	Information Dissemination	Share mission information with external users and systems
CAP-330	TAK Integration	Publish position, observations, tracks, and Cursor-on-Target events to TAK environments
CAP-340	Mission Alerting	Notify operators of mission-critical events, system warnings, and operational anomalies.

7.4 Tracking and Mapping Capabilities

Capability ID	Capability	Description
CAP-400	Target Tracking	Maintain awareness of designated targets and points of interest
CAP-410	Target Observation	Observe and characterize tracked targets
CAP-420	Track	Maintain target position and movement

	Maintenance	information
CAP-430	Mapping Support	Support area characterization and mapping activities

7.5 Autonomous Operations Capabilities

Capability ID	Capability	Description
CAP-500	Autonomous Operations	Execute predefined behaviors with limited operator intervention
CAP-510	GPS-Degraded Operation	Support operation in environments with limited GPS availability
CAP-520	Lost Link Recovery	Detect communication failures and initiate recovery actions
CAP-530	Low Battery Recovery	Detect low battery conditions and initiate recovery procedures
CAP-540	Autonomous Docking and Charging	Support autonomous recovery and battery replenishment

7.6 Mission Data Management Capabilities

Capability ID	Capability	Description
CAP-600	Mission Data Management	Manage mission products and historical mission information
CAP-610	Mission Data Archiving	Store mission data following mission completion
CAP-620	Mission Data Retrieval	Retrieve historical mission products
CAP-630	Mission Traceability	Maintain mission logs, telemetry, and event records
CAP-640	Mission Data Export	Export mission products, imagery, telemetry, and logs for external analysis and reporting.

7.7 System Management Capabilities

Capability ID	Capability	Description
CAP-700	System Health Management	Monitor subsystem status, detect faults, and report system health information to the operator.

8. Operational Modes

Mode ID	Mode Name	Description
OM-001	Standby	System powered off or inactive while awaiting deployment.
OM-002	Pre-Mission	System initialization, configuration, health checks, and mission setup activities.
OM-003	Launch	Airborne platform departs the launch location and transitions to mission execution.
OM-004	Mission Execution	Primary ISR operations including observation, reconnaissance, mapping, and information collection.
OM-005	Tracking	Active target tracking and continuous observation of designated objects or areas of interest.
OM-006	Mapping	Collection and generation of area characterization or mapping products.
OM-007	GPS-Degraded Operation	System operates with reduced or unavailable GPS services using alternative navigation methods.
OM-008	Lost Link Recovery	Autonomous recovery actions initiated following loss of communications.
OM-009	Low Battery Recovery	Autonomous recovery actions initiated due to low battery conditions.
OM-010	Return to Operator	System navigates toward the operator or designated recovery location.
OM-011	Return to Charge	System navigates toward an autonomous charging station.
OM-012	Docking	Autonomous alignment and docking with charging infrastructure.
OM-013	Charging	Battery recharge and readiness monitoring activities.
OM-014	Emergency Landing	Controlled landing due to fault conditions or safety concerns.
OM-015	Shutdown	Mission termination and system power-down activities.

9. Measures of Effectiveness and Success Criteria

9.1 Measures of Effectiveness (MoEs)

ID	Measure of Effectiveness
MOE-001	Improve operator situational awareness of the area of interest
MOE-002	Reduce need for personnel to physically enter hazardous environments
MOE-003	Provide timely operational information to support decision-making
MOE-004	Successfully identify and observe points of interest
MOE-005	Successfully maintain awareness of designated targets during tracking operations
MOE-006	Enable mission execution in GPS-degraded or GPS-denied environments
MOE-007	Provide usable mission products for post-mission review and analysis

9.2 Measures of Performance

ID	Measure of Performance
MOP-001	Mission endurance (minutes)
MOP-002	Time required to deploy system
MOP-003	Communications range
MOP-004	Video transmission latency
MOP-005	Target tracking accuracy
MOP-006	Mapping coverage area
MOP-007	Lost-link recovery success rate
MOP-008	Autonomous docking success rate
MOP-009	Time required to achieve autonomous recharge
MOP-010	Mission archive completion rate

9.3 Success Criteria

ID	Success Criterion
SC-001	TOMTOM successfully deploys and launches from a portable configuration
SC-002	TOMTOM collects and displays mission-relevant information to the operator
SC-003	TOMTOM supports observation, tracking, or mapping activities within the area of interest
SC-004	TOMTOM successfully archives mission data for post-mission review
SC-005	TOMTOM successfully executes predefined recovery actions following lost-link events

SC-006	TOMTOM successfully executes predefined low-battery recovery procedures
SC-007	TOMTOM demonstrates operation in both indoor and outdoor environments
SC-008	TOMTOM demonstrates operation under GPS-degraded or GPS-denied conditions
SC-009	TOMTOM demonstrates integration with ATAK/TAK-compatible systems
SC-010	TOMTOM demonstrates autonomous charging or battery recovery capability

Appendix A – Supporting Diagrams

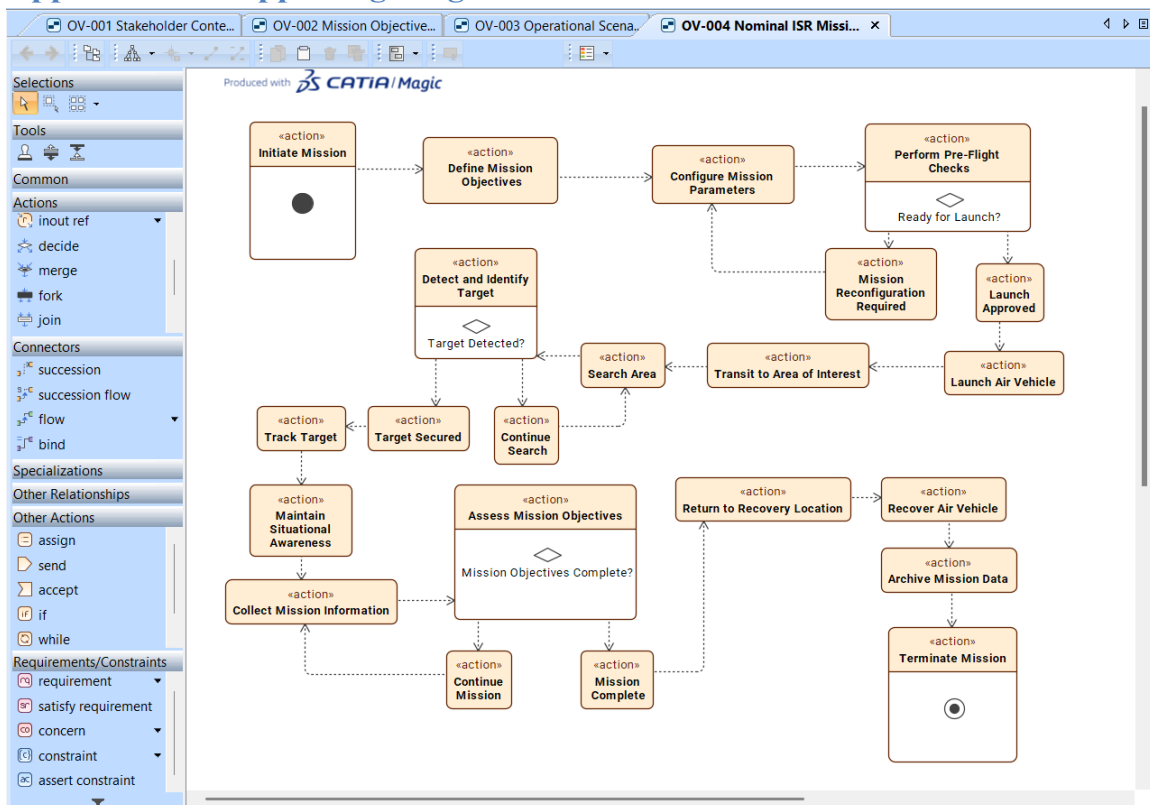


Figure 1. Mission Threads Diagram.

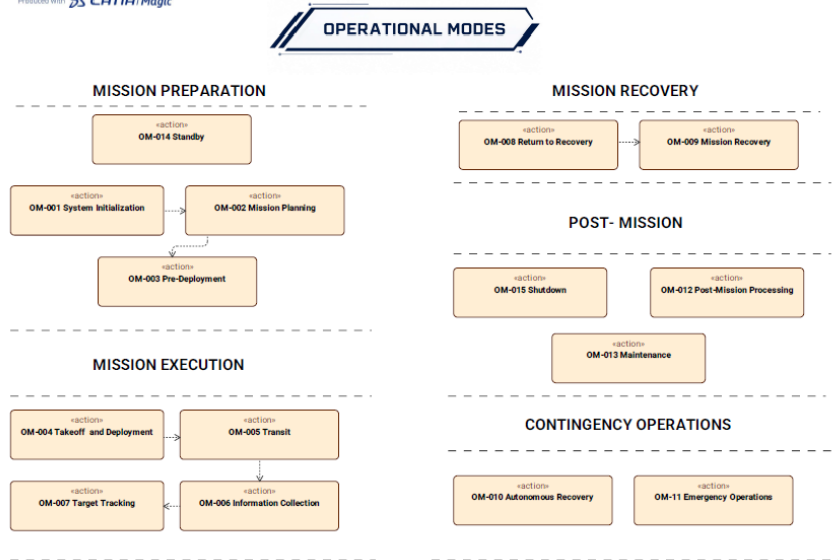


Figure 2. *Operational Modes diagram.*